

1.

What is the worst-case time complexity to insert an element at the end of a singly linked list, given a head pointer only?

- A. $O(n)$**
- B. $O(1)$**
- C. $O(2n)$**
- D. $O(\log n)$**

Answer: A

2.

Which of the following is a disadvantage of using a singly linked list compared to an array?

- A. Fixed size of data can be added**
- B. Facing difficulty in managing memory.**
- C. Provide inefficient random access.**
- D. Has Limited storage capacity**

Answer: C

3.

Which of the following is NOT considered a fundamental operation directly supported by a singly linked list?

- A. add new element at the beginning**
- B. Deletion last element**
- C. Print list**
- D. Print element from given index**

Answer: D

4.

```
int fun(Node* head) {  
    Node* slow = head;  
    Node* fast = head;  
    while (fast != null && fast->next != null)  
    {  
        slow = slow->next;  
        fast = fast->next->next;  
    }  
    return slow->data;  
}
```

If the linked list is 767 -> 676 -> 766 -> 776 -> 777 -> NULL
before calling fun, what will be the return value from fun call?

- A. 767
- B. 766
- C. 776
- D. 777

Answer: B

5.

Which of the following concepts is not directly applicable or supported by the structure of a singly linked list?

- A. Dynamic size
- B. Storing data elements
- C. Searching data element
- D. Bidirectional traversal

Answer: D

6. Which of the following is the type of linked list?

- A. Header Linked List
- B. Singly linked list
- C. Self-Organizing List
- D. Skip List

Answer: B